



(16-05-24)

End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 239 2029

Name of the Program and semester: BCA II

Name of the Course : Fundamentals of python programming

Course Code: TBC 211

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

- a. What are data types in Python? Explain numeric, sequence, and mapping types with examples. CO1
- b. Explain the important features of Python and mention five areas where Python is commonly used. CO2
- c. i. Write a program to calculate the sum of digits of an input number using while loop. CO2
ii. Write a program to check whether number is Armstrong number or not using for loop.

Q2. (2X10=20 Marks)

- a. Explain the range() function in with its syntax, parameters, and usage in loops. Also write a program to print all even numbers between 1 and N using the range(). CO3
- b. What is event-driven programming? Explain how buttons, labels, and entry fields are used in GUI programs using tkinter. CO4
- c. Explain various types of conditional statements in Python with syntax and suitable examples. CO1

Q3. (2X10=20 Marks)

- a. WAP to print the area of circle, rectangle and square by using function with return type. CO1
- b. Explain different types of arguments used in function. Also write a program to implement *args and **kwargs. CO3
- c. What is a dictionary in Python? Explain how to create, access, modify, and iterate through a dictionary with examples. CO3

Q4. (2X10=20 Marks)

- a. Define the following with example: CO4
 - i. Self parameter
 - ii. Lambda function with map()
 - iii. init__ method
- b. What is recursion? Write a program to print the sum of digit of an input number using recursive function. CO3
- c. Explain file handling in Python. Describe various file modes and methods used for reading and writing text files. CO4

Q5. (2X10=20 Marks)

- a. Explain the concept of a set in Python. What are its key properties and operations? Give examples. CO5
- b. Explain indexing, negative indexing and slicing in string with example. What are f-strings in Python? Show how they are used to format strings with variables. CO5
- c. i. Write a Python program to read a text file and count the total number of lines, words, and characters.
ii. Create a GUI-based program using tkinter to input two numbers and display their sum when a button is clicked. CO4